rail_damper_3.csv
(a) transverse position error |p - p_gamma(eta)|
mm
t (s)
safety stop
robot
preflight (robot's model)
|xi| bound if all position (40 mm)
(b) transverse attitude error |log(R_gamma(eta)^T R)|
deg
t (s)
safety stop
robot
preflight (robot's model)
|xi| bound if all rotation (22.9 deg)
(c) transverse error |xi| (both, as the safety check sees it)
|xi| (metric units)
t (s)
safety stop
robot
preflight (robot's model)
bound 0.04
(d) redundancy output beta
beta (rad)
t (s)
safety stop
robot
preflight (robot's model)
bound 0.05 rad
(e) path speed eta_dot against nu_d
eta_dot (metric units / s)
t (s)
safety stop
robot
preflight (robot's model)
nu_d
(f) delivered torque |tau_i + G_i| / tau_max,i
fraction of the limit
t (s)
safety stop
joint 1
joint 2
joint 3
joint 4
joint 5
joint 6
joint 7
bound 60 %
(g) distance of each joint to its nearer limit
rad
t (s)
safety stop
joint 1
joint 2
joint 3
joint 4
joint 5
joint 6
joint 7
bound 0.10 rad
(h) max |qd_i|/qd_max,i and commanded |qdd*_i|/qdd_max,i
fraction of the limit
t (s)
safety stop
|qd| (robot)
|qdd*| (robot)
|qdd*| (preflight (robot's model))
bounds 50 % and 30 %